# Pose Estimation and Virtual Gym Assistant using MediaPipe and Machine Learning

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***Abstract*—This research paper demonstrates a virtual fitness trainer, using Google's Mediapipe library, which was made for various multimodal machine learning and deep learning pipelines. By monitoring the user's every step with posture estimation algorithms, the suggested system can provide useful feedback and insights as they work out. Mediapipe's deep algorithms and posture estimation module are utilized to develop this novel system, which captures user movements by identifying unique body landmarks for each exercise. The system also keeps track of the total amount of reps quite precisely. The system determines the user's performance by computing angles and landmarks and feeding this data into a machine-learning model, which then classifies the correct posture and the number of repetitions of the exercise. The system uses many fitness datasets in addition to user-supplied data for thorough training and assessment. A detailed comparison of machine learning models like Logistic Regression, Support Vector Machine, Naive Bayes, Decision Tree, and Artificial Neural Network is included in this research. The findings prove that our suggested method provides accurate and individualized feedback, which improves exercise performance and lessens the likelihood of damage.**

***Keywords— Mediapipe, Computer Vision, Pose Estimation, Fitness, Exercises, Deep Learning, Machine Learning***

## I. Introduction

Physical activity is essential in today's society since it helps build stronger bones and muscles and enhances your capacity to carry out daily tasks. People today are more health- conscious and exercise daily, either at home or at the gym. People exercised at home during the COVID-19 pandemic. But this presents some issues. Injuries and other issues result from people not knowing how to work out. People also choose home workouts since gyms are expensive.

This study will show how to exercise at home without going to the gym every day. The number of reps performed were counted by identifying human body landmarks such shoulders, elbows, knees, wrists, ankles, etc. using the Mediapipe library, which Google developed. This paper includes rep counts for popular workouts like bicep curls, squats, front kicks, and jabs. Biomechanics is extremely indispensable for our topic as the paper aims to estimate the correct posture of a particular exercise. This paper also uses machine learning and deep learning to assess pose accuracy and exercise technique and com- pares machine-learning accuracy. Unique datasets are used to test strategies with and without hyperparameter adjustment. Artificial Neural Network (ANN) is employed to calculate accuracy and compared all methods to find the best for this project.

The following sections are provided in chronological order in the paper: Literature Survey, Proposed Methodology, Algorithm Summary, Dataset, Implementation, Results and Discussion, Conclusion and Future Scope, and References.

## II. Literature Survey

Mediapipe-based pose estimation research is an emerging field. Arpita Halder et al. compared SVM, KNN, and other ma- chine learning models to detect vernacular American, Indian, and Italian sign languages. The Mediapipe Palm Detection Model is used for hand image data. They propose landmark extraction, cleaning, normalization, and splitting for training and testing. Machine learning models use preprocessed data. According to the results, SVM outperformed other algorithms [1]. In a paper (A. Singh et al.), the challenges surrounding estimating a person's pose are covered, along with an overview of significant pose estimation research, covering both deep learning methods and conventional image-based approaches. The authors have created a straightforward model using a convolutional neural network that estimates the postures and exemplifies the potential of CNNs after examining numerous findings and identifying the constraints [2]. Several works focus solely on yoga and gym workouts. The writers of Chen, Steven et al. built a gym exercises assistant model employing two methods. The first method sets geometric heuristics of threshold values manually using logical deductions. The second machine learning strategy uses the closest neighbor classifier and dynamic time warping. OpenPose, which estimates posture using part affinity fields is used for pose estimation. Vector fields encode limb location and orientation [3].

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Agarwal et al. briefly describe fitness-related AI applications. GOQii, Cure. fit, HealthifyMe, and others offer gym alternatives and promote fitness on smartphones. The authors suggested using machine learning to evaluate asanas. Mediapipe's BlazePose model detects postures by identifying and analyzing 33 pairs of critical spots on the human body. A pose-analyzing tool calculates landmark angles heuristically [4]. X. Li, M. Zhang, et al. used Mediapipe to identify basic fitness movements. Up and down detection states are divided into picture samples in the first of three fitness action recogni- tion algorithm phases. KNN-classifies fitness actions using the training set. They calculate the recognition angle and threshold from categorization findings by counting fitness activities. The Mediapipe Blazepose model is superior for customized AI fitness on mobile devices than Openpose, Alphapose, etc. [5]. A paper (by Kumar D. et al.) proposes an android application for yoga pose estimation using the OpenPose model for key landmark detection. The pipeline proposed uses a dataset of yoga poses which are preprocessed using OpenPose and stored on the local machine and then the real-time captured event of the pose is compared with the local machine data to determine the pose [6].

G. Taware et al. used occlusion-simulating augmentation to test the posture detector in strong occlusions, which are different from typical tests. Before employing user alignment, the estimator locates the user's 33 critical points. It uses regression and heatmaps. The test model's layers are trimmed after using the training model's algorithms. The heatmap was also used by the authors to analyze the encoder's lightweight integration [7]. A paper by Kanase et al. presents a method for pose estimation using the pre-trained OpenPose model, a multistage CNN model. This model finds key points in videos. Key points are normalized and cleaned. The authors suggest using a heuristic method and machine learning for pose estimation. The heuristic method uses the OpenPose model's landmark angles. The model uses dynamic time warping to classify videos. Euclidean distance between key points is computed and represented in a distance matrix. Best match points are chosen, and distance is calculated.[8]. In a paper, Anilkumar et al. used the BlazePose model by Mediapipe to obtain body joint coordinates. Coordinates are analyzed and compared with yoga position data. Users can set the flexibility threshold variable as needed. Feedback is given to users based on results [9].

Another paper also describes this same approach for AI-fitness repetition capabilities. The authors used the BlazePose model to get the 33 landmark coordinates and then computed the angles of the required joints and compared them to the geometric heuristics of the exercise [10]. In another study (Chen KY et al.), a fitness database was created using deep transfer learning. A neural network was trained to identify fitness movements. Researchers used Yolov4 and Mediapipe for quick fitness movement detection. They captured the fitness signal of each movement to assemble the database. MLP was used for classifying fitness signal waveform. Fitness and completeness classifications had positive outcomes [11]. A system (Pauzi, A.S.B. et al.) monitors body activity and adds labelled skeleton joints to a video. This solution uses deep learning for sports and physically demanding jobs with PoseNet dataset and Mediapipe Blazepose algorithm. Accuracy is found within 10% of IMU-based motion capture. The system labels and calculates joint movement velocity and angles [12].

Ashish Ohri et al. showed how to implement real-time pose estimation and correction in a user-friendly app. OpenPose, Pifpaf, Tensorflow-Lite, and TensorflowJS are some posture estimation models used for position correction. Few models are fast and accurate, and model-selection depends on the application [13]. Long, C et al. developed a yoga self-coaching system using transfer learning. The study started by collecting yoga posture data using a webcam, then using data aug- mentation techniques. Study analyzed transfer learning using MobileNet model. They created an AI yoga system with a real- time prediction model. The yoga posture classification method had a 98.43% accuracy rate in our self-coaching system. It predicts outcomes and gives guidance for incorrect postures. Incorrect posture is detected using the Mediapipe algorithm for joint angle calculation [14].R. Gadhiya et al. analyzed pose estimation models like OpenPose, BlazePose, DeepPose, and Hourglass to discuss their pros and cons in different scenarios. The analysis included push-up exercise with OpenPose and BlazePose. Results were compared and analyzed [15].

## III. Proposed Methodology

The complete process pipeline is separated into two stages which are described below.

### *A. Landmark Detection Stage*

The subject is required to place themself in front of the camera so that the regions of interest are visible to it. With the help of Mediapipe, we first detect the subject/region of interest(ROI) [15]. Using Mediapipe's Pose Landmark Model, we detect the landmarks of various body parts, which are discussed later.

### *B. Rep Counting Stage*

- The landmarks of a particular frame are extracted and the angles between the desired reference landmarks are calculated. The inputs are then passed to a threshold function based on the angle heuristic, which is different for each exercise. If the angle extracted satisfies the threshold, the rep count is incremented by 1; otherwise, it is incremented by 0.
- If the angles satisfy the threshold, the extracted landmarks are fed to a machine learning model which classifies the form of the pose to determine how accurately the exercise is being performed. If the confidence score is above 60 percent for the correct form, the system validates the current rep and increments the rep count. Otherwise, it increments it by 0 and continues. To obtain more accurate results of the correct pose, the user has implemented and compared various ML models

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and DL models for the form classification.

### C. *Pose Detection and Angle calculation*

- Pose detection makes it easier to keep track of the posture during any exercise set. Pose detection models have a set of predefined landmarks which help in identifying various body parts easily.

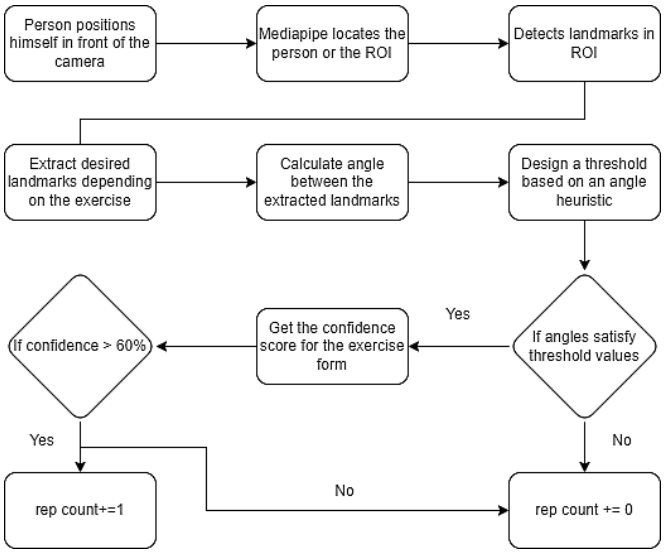


Fig.1. Process Pipeline

Some examples of pose detection models include PoseNet, AlphaPose, OpenPose, and many others. After some research, BlazePose was the model selected for this proposed application. Fig. 1 illustrates the pipeline of the proposed posture estimation, rep counting, and angle calculation model based on heuristic values which are explained further in the paper in detail.

*1) The BlazePose Model:* Mediapipe's pose detection API uses BlazePose. The BlazePose model performs slightly poorer than OpenPose but is suitable for Yoga/Fitness positions. The Pose Detection API detects landmarks in the proposed system. BlazePose's landmark coordinates are shown in Fig.2. It is a human pose estimation model developed by Google Research that can detect 33 key points of the human body, including the head, torso, arms, and legs. BlazePose is fast and accurate, making it ideal for real-time fitness tracking, virtual try-on, and gaming. BlazePose uses a lightweight neural net- work architecture with convolutional and depthwise separable convolutional layers for high accuracy and low computing complexity. The model learns from a huge dataset of annotated photos using supervised and self-supervised learning. We will extract landmarks to calculate angles for specific exercises, then predict the accuracy, and count reps [16].

Mediapipe's pose detection API implements the BlazePose model. The BlazePose model performs slightly worse compared to the OpenPose model, but it is an ideal choice for Yoga/Fitness poses. The proposed system uses the Pose Detection API for landmark detection. Fig. 2 below depicts the landmark coordinates from BlazePose's pose and landmark detector model which provides human pose tracking. It is a human pose estimation model developed by Google Research that can detect 33 key points of the human body, including the head, torso, arms, and legs.

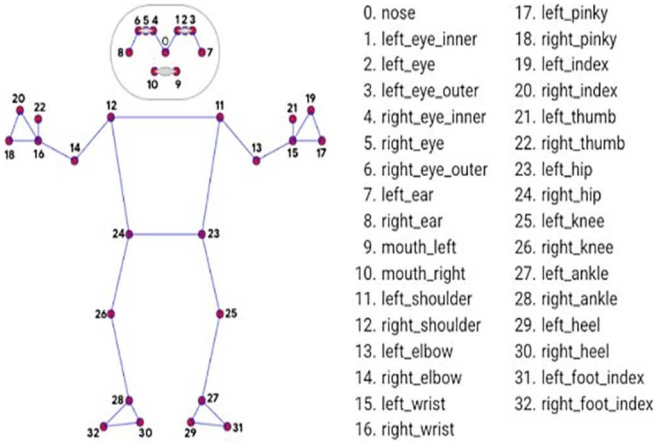


Fig.2. Landmarks of Mediapipe's Blaze Pose Model [16]

*2) Algorithm used for detecting Squats:* Squat analysis can be performed by dividing motion into three complete domains. Upper body, lower body, and movement mechanics. The lower body evaluates the alignment of the hip, knee, and ankle landmarks, the upper body focuses on stability and posture of the head, neck, and stomach core, and movement mechanics evaluate workout time and coordination. The shoulder, knee, hip, ankle, and foot joints and their spatial angles are used for posture or attitude calculation. The back and the thigh landmarks calculate the hip angle. The thigh and the lower leg landmarks calculate the knee angle. The landmark coordinates are extracted from BlazePose's pose and landmark detector model which provides human pose tracking of 33 landmarks [16][17]. Given below is the heuristic value of the various angles used to check for a correct and accurate squat posture. Fig. 3 below shows the squats exercise being done and the desired landmarks being tracked by MediaPipe in real-time.

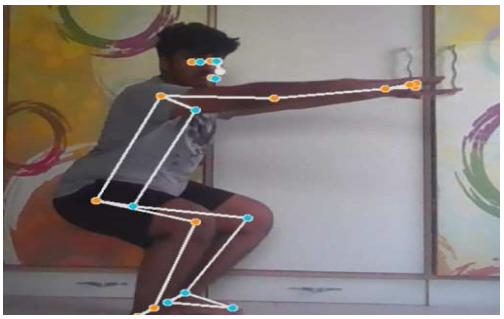

Fig.3.Squats exercise implementation

TABLE I. TABLE OF THRESHOLD HEURISTIC ANGLE VALUES FOR SQUATS EXERCISE

| Angles | Heuristic Values | Remarks |
|---|---|---|
| Hip Angle | 50-71 degrees | less than 44 degrees will cause imbalance |
| Knee Angle | 55-68 degrees | greater than 75 degrees will reduce the effect of the rep |

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| | | |
|---|---|---|
| Torso Angle | 35-43 degrees | greater than 45 will lead to incorrect posture |
| Ankle Angle | 75-85 degrees | less than 75 will cause too much strain on the ankles |

*3) Algorithm used for detecting Bicep curls:* Two heuristics are crucial for estimating bicep angle pose [3]. When the angle between the upper arm and torso is greater than 30 degrees, the curl is inaccurate. Lifting big weights that the body cannot handle causes excessive upper arm rotation [2].

Initially, the angle between the forearm and the upper arm would be 150-180 degrees. This angle decreases when the forearm goes all the way up and the bicep muscles contract and increases when it goes back to its initial position and the muscles relax. If the angle between the upper arm and the forearm is greater than 70 degrees during contraction, the curl is improper, and thus by using these measures and threshold values we count the number of reps and predict the accuracy accordingly [3]. So we use the shoulder, hip, elbow, and wrist joints for the bicep curl pose estimation. Fig. 4 below illustrates the closed and open angles of the arms during a bicep curl rep with the landmarks tracked by MediPipe.

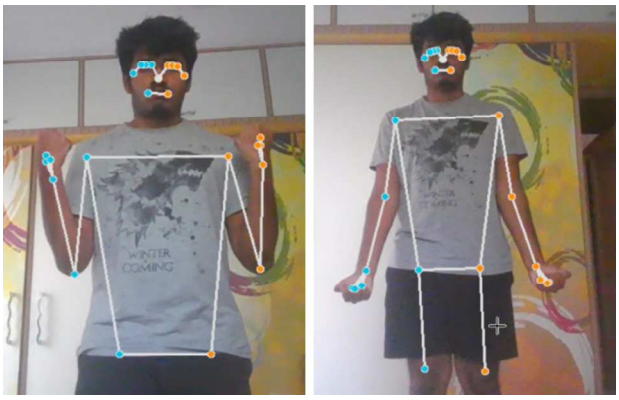

Fig. 4. Bicep curls exercise Implementation (Closed arms and Open arms respectively)

TABLE II. TABLE OF THRESHOLD HEURISTIC ANGLE VALUES FOR BICEP CURLS EXERCISE

| Angles | Heuristic Values | Remarks |
|---|---|---|
| formed by torso and upper arm | less than 35 degrees | greater than 45 degrees will result in incorrect rep |
| Formed by upper arm and forearm during contraction of bicep | less than 70 degrees | greater than 70 degrees means incomplete rep |

*4) Algorithm used for detecting Boxing-jabs:* The jab punch exercise's two major principles are hip rotation and upper-body weight transfer. Punching should not bend the arms. Reps are calculated using both arms' shoulders, elbows, wrists, and the hip. Shoulder, elbow, and wrist joints determine the initial angle. The hip-shoulder-elbow angle is particularly crucial in jab punching [18]. The arm angle is the wrist, shoulder, and elbow angle, whereas the hip-rotate angle is the hip, shoulder, and elbow angle. Two stages—in and out—are also discussed.

TABLE III. TABLE OF THRESHOLD HEURISTIC ANGLE VALUES FOR BOXING- JAB PUNCH EXERCISE

| Angles | Heuristic Values | Remarks |
|---|---|---|
| Arm Angle(shoulder-elbow-wrist) | 150-180 degrees | less than 140 will generate lesser force |
| Hip-Rotate Angle(hip-shoulder-elbow) | 80-100 degrees | less than 65 degrees will work the arms more |

*5) Algorithm used for detecting Front Kicks:* This exercise will focus on two heuristic angles like the others. Ankle, knee, hip, and shoulder joints are utilized. The ankle-knee-hip angle (leg angle) and shoulder-hip-knee angle (hip angle) are of importance. Two stages are considered while performing one rep of a front kick – the 'in' and 'out' stages. The 'in' stage is the beginning, and the 'out' stage is when the kick happens and one leg moves away from the body. Leg and hip angles should be fewer than 50 degrees and 70 degrees during the "in" stage. The table below shows the heuristic values for a correct front kick during the 'out' stage.

TABLE IV. TABLE OF THRESHOLD HEURISTIC ANGLE

| Angles | Heuristic Values | Remarks |
|---|---|---|
| Leg angle(hip-knee-ankle) | more than 120 degrees | less than 110 degrees will result in incorrect posture |
| Hip angle(shoulder-hip-knee) | 71-120 degrees | more than 130 degrees will work the back muscles more |

VALUES FOR FRONT KICKS EXERCISE

### *A. Form Classification*

The angle heuristic may work in some circumstances, but it's important to watch the subject's form during each exercise. Thus, if the threshold condition occurs, the application tests the subject's form by sending the current landmarks to a model trained on the exercise's acceptable postures at standard rep counts. The application validates the rep and increments the rep count if the model classifies the current form into one of the valid posture labels with a confidence score of at least 0.6. Only bicep curls were implemented. Different models were trained on images of persons performing bicep curls at the endpoints of a rep—an open arm position (labeled "1") and a closed arm position (marked "0").

*1) Machine Learning Approach:* Four different machine learning models were used for the research: Support Vector Machine, Logistic Regression, Naive Bayes Classifier, and Decision Tree Classifier. These classification models were employed to compare the results and find the best classifier for the dataset. The models were used with and without hyperparameter tuning to understand the impact of the latter and enhance classification efficiency.

*a) Support Vector Machines:* Support vector machines (SVMs) use labeled training data to construct functions for mapping input-output pairs. The SVM generates maximum-margin hyperplanes and depends only on a subset of the training data that are near the class bound- aries to produce the final model. The SVM equation is given by Eq. (1):

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$$h(x) = sign\left(\sum_{i=1}^{nSV} \alpha_i\, y_i \cdot K(x, x_i) + b\right) \quad (1)$$

*b) Logistic Regression:* Logistic regression employs a sigmoid function, also known as a logistic function, to transform predictions into their corresponding probabilities. Eq. (2) below represents logistic regression:

$$P\,(y\ =\ 1\mid \mathbf{x})\ =\ \frac{1}{1+e^{-(w^T x+b)}} \quad (2)$$

*c) Naive Bayes Classification:* Naive Bayes is a machine learning algorithm that uses the Bayes Theorem to classify data by assigning probabilities to different categories based on the features of the data. Eq. (3) .

$$P(y|x) = \frac{P(x|y)\cdot P(y)}{P(x)} \quad (3)$$

*d) Decision Tree Classifier:* Decision Trees use a set of rules or criteria that typically rely on one variable at a time. They divide the input data into distinct regions, gradually increasing the level of detail at each step until they reach the final predicted label at the end of the tree. The Gini index is a measure of impurity that is commonly used in Decision Trees for binary classification problems. Eq. (4) illustrates the formula for the same.

$$Gini = 1 - \sum_{i=1}^{c} {p_i}^2 \quad (4)$$

*2) Deep Learning Approach:* For the deep learning approach, an Artificial neural network (ANN) was used. In the ANN model, there are several dense layers of several neurons. The model has a 32-neuron input layer followed by a 16- neuron hidden layer and a single neuron output layer. The first and second layers have a ReLU activation function while the last layer has a sigmoidal activation function. The proposed model uses binary cross entropy as the loss function. It was also observed that using more than 3 layers causes the network to overfit. Eqs. (5) and (6) showcase the formulae for sigmoid and RELU activation respectively where x is the input to the function.

$$Sigmoid(x) = \frac{1}{1+e^{-x}} \quad (5)$$

$$Relu(x) = \max(0, x) \quad (6)$$

A new dataset was required to implement pose detection specifically for the bicep curls workout. The dataset used was sourced from Kaggle and provided by Jiunn [19], which was freely accessible. Despite having data for three exercises, the focus was on bicep curls. The initial data was short videos, but pose detection in bicep curls required image data with the correct pose. Data was extracted from the Kaggle dataset to generate the appropriate images. Frames from the movies that met the angle heuristic threshold condition for bicep curls were chosen. Specifically, frames with the arm in the open position(between 150 and 180 degrees) and closed position (less than 70 degrees) were considered.

Another dataset uploaded by Abdillah on Kaggle [20] was utilized. Pose detection images were carefully reviewed. The same angle heuristic threshold condition was used to guarantee the photographs captured the appropriate bicep curl poses. Despite efforts, the dataset was limited. Bicep curl stance photos were added to the dataset. These photos were merged with earlier datasets. Mediapipe's Pose Landmark Model extracted landmarks from the photos, reflecting posture detection's essential places. The dataset was CSV-formatted landmark data. The bicep curls dataset had 197 samples, containing 100 closed-arm postures and 97 open-arm postures. This balanced the classes for accurate machine learning model training.

TABLE V. HYPERPARAMETER TUNING

| **Models** | **Hyperparameters Tuned and Tuned Value** |
|---|---|
| Support Vector Machine | • Regularization parameter(inverse)= 10<br>• Kernel coefficient = 0.01<br>• Kernel type kernel = poly |
| Logistic Regression | • Regularization strength = 100.0<br>• Kernel Penalty = L1<br>• Kernel Algorithm = liblinear |
| Naive Bayes Classifier | • Variance smoothing = 1.0 |
| Decision Tree Classifier | • criterion = gini<br>• max-leaf nodes<br>• min samples split = 8<br>• min samples leaf = 1<br>• max depth = 7 |
| Artificial Neural Network | • layers = [32,16]<br>• batch sizes = 8<br>• activation functions = ReLU<br>• epochs = 15 |

After using the default parameters for the four machine learning models, we tuned the hyperparameters for the same as given in Table V by using GridSearchCV to find the optimum parameters. GridSearchCV also helps us to perform cross- validation, for which we set the value to 5 folds for each model. For the deep learning approach using an Artificial Neural Network, the model was further also tuned using the GridSearchCV algorithm for different layers, batch sizes, activation functions, and epochs.

## IV. Results and Discussion

The experiment results are summarised in Tables 6, and 7. Tables VI uses abbreviations for the following classifiers: SVC (Support Vector Classifier), LR (Logistic

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Regression), NBC (Naive Bayes Classifier), and D-Tree (Decision Tree Classifier). The machine learning models were trained on 137 training images and evaluated on 60 test images. For models tuned with GridSearchCV, cross-validation was conducted with 5 folds. The accuracies of most models, except for SVC, have improved after parameter tuning. Notably, the Na¨ıve Bayes classifier has shown a significant increase in accuracy. The Na¨ıve Bayes classifier experienced a statistically significant 17% improvement following parameter tuning. The accuracy of SVC did not change, while the increase in accuracy for Logistic Regression and Decision Tree classifier was minimal. After tuning the hyperparameters, the accuracy of the Artificial Neural Network (ANN) decreased by 3 percent, resulting in a final accuracy of 90 percent. The Mean Squared Error has decreased to 0.0507. The time and effort invested in hyperparameter tuning for ANN model does not justify the marginal improvement in MSE. Logistic Regression demonstrates superior accuracy and simplicity in comparison to alternative models following hyperparameter tuning. The logistic regression model achieves a 99% accuracy rate.

The findings demonstrate a noteworthy enhancement in the development of an AI gym application system compared to previous methods. The DTW classification model in reference [4] achieved an 89% accuracy for the bicep-curl exercise. The dataset in [4] was a video dataset, while the dataset in this research is an image dataset. One limitation of this study is the small dataset size. To improve the results, it is recommended to increase the dataset size and obtain more precise image representations of the exercise. In contrast to the methodology proposed by [7], this study presents a simpler logistic regression approach within its architectural framework. This approach is believed to offer more advantages when compared to a complex neural network architecture. Therefore, the computational burden on the server is reduced. The Non-Maximum Suppression algorithm has limitations when applied to certain human postures. These limitations could be addressed by using a more generalised approach. A hybrid architecture was proposed [11] that combines YOLO V4, MLP classificationfor fitness 1D signal waveforms, and Mediapipe. It achieved an accuracy of 98.56% for YOLO V4 and 92.84% for MLP classification. Although the study used a custom dataset similar to this research, the architectural design differs significantly.

### *A. Requirements*

All the experimentation was performed on a machine with the following specifications:

*System Requirement: Operating System:* Windows 11, CPU: Intel(R) Core(TM) i7-10750H CPU @ 2.60GHz, RAM: 24 GB, and GPU: NVIDIA GeForce GTX 1650 Ti.

*Software Requirement:* python 3.9.5, mediapipe 0.8.9.1, opencv-python 4.5.5.62, pandas 1.4.1, and scikit-learn 1.0.2

TABLE VI. RESULTS OF ML MODELS

| Sl. No. | Model | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|---|
| 1 | SVC | 0.98 | 0.98 | 0.99 | 0.98 |
| 2 | LR | 0.96 | 0.96 | 0.98 | 0.97 |
| 3 | NBC | 0.77 | 0.86 | 0.72 | 0.72 |
| 4 | D-Tree | 0.97 | 0.97 | 0.96 | 0.97 |
| 5 | SVC (tuned) | 0.98 | 0.98 | 0.99 | 0.98 |
| 6 | LR (tuned) | 0.99 | 0.98 | 0.97 | 0.99 |
| 7 | NBC (tuned) | 0.94 | 0.94 | 0.96 | 0.95 |
| 8 | D-Tree (tuned) | 0.96 | 0.96 | 0.98 | 0.97 |

TABLE VII. ACCURACY AND MSE OVERVIEW OF ANN MODEL

| | Without hyperparameter tuning | With hyperparameter tuning |
|---|---|---|
| MSE | 0.0623 | 0.0507 |
| Accuracy | 93 | 90 |

## V. Conclusion and Future Scope

The proposed application monitors body movement during exercise and offers feedback on posture. To accurately evaluate MediaPipe, it is important to recognise its limitations. MediaPipe faces challenges when operating in densely populated environments. Pose estimates can be inaccurate in group exercise due to variations in body parts and occlusions. This may lead to misinterpretation of the body's position and result in inaccurate user feedback. Occluded body parts pose a challenge for MediaPipe's pose calculation. The system's ability to accurately assess user feedback may be compromised when body components are obscured, leading to difficulties in determining the user's stance.Using contextual information or multi-person pose estimation to enhance pose estimation in crowded scenarios has potential benefits. These situations could benefit from utilising methods that leverage the presence of individuals in the surrounding environment or the contextual factors at play. Depth sensors or RGB-D cameras can assist in detecting body posture in obscured areas in challenging environments. Temporal information can be utilised by researchers to infer the positions of missing body parts by analysing the trajectories of visible body parts.

Various machine-learning techniques were used to estimate exercise posture accuracy for bicep curls, front kicks, boxing (jab punches), and squats. The study uses MediaPipe library to detect body posture and generate exercise angles using the built-in pose estimator model. Heuristic functions helped determine exercise thresholds for identifying correct repetitions. The models were then fed the data to predict the state of the rep for a given exercise. The best results were given by SVM without hyperparameter tuning and Logistic Regression with hyperparameter tuning. An ANN was also given the same data and produced good results. Theis study presented a comprehensive analysis of pose estimation, encompassing widely-used methodologies and assessment metrics.

Due to limited resources and time, certain other unique features could not be completed but are part of this application's future scope. Expanding the dataset and adding data improves algorithms and models. The idea is to provide

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an application with a variety of activities for everyone. Users can access gamification, customised workouts, and diet plans. The goal is a healthcare software with pose estimation, individualized nutrition and training routines, and gamification. VR/AR is planned for an engaging user experience. People with limited gym time can benefit from AR/VR home automation.

## REFERENCES


[1] Halder, Arpita, and Akshit Tayade. "Real-time vernacular sign language recognition using mediapipe and machine learning." Journal homepage: www. ijrpr. com ISSN 2582 (2021): 7421.

[2] A. Singh, S. Agarwal, P. Nagrath, A. Saxena and N. Thakur, "Human Pose Estimation Using Convolutional Neural Networks," 2019 Amity International Conference on Artificial Intelligence (AICAI), Dubai, United Arab Emirates, 2019, pp. 946-952, doi: 10.1109/AICAI.2019.8701267.

[3] Chen, S., Yang, R. R. (2020). Pose trainer: correcting exercise posture using pose estimation. arXiv preprint arXiv:2006.11718.

[4] V. Agarwal, K. Sharma and A. K. Rajpoot, "AI based Yoga Trainer - Simplifying home yoga using mediapipe and video streaming," 2022 3rd International Conference for Emerging Technology (INCET), 2022, pp. 1-5, doi: 10.1109/INCET54531.2022.9824332.

[5] X. Li, M. Zhang, J. Gu and Z. Zhang, "Fitness Action Counting Based on MediaPipe," 2022 15th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI), Beijing, China, 2022, pp. 1-7, doi: 10.1109/CISP-BMEI56279.2022.9980337.

[6] Kumar, D., Sinha, A. (2020). Yoga pose detection and classification using deep learning. LAP LAMBERT Academic Publishing.

[7] G. Taware, R. Kharat, P. Dhende, P. Jondhalekar and R. Agrawal, "AI-Based Workout Assistant and Fitness Guide," 2022 6th International Conference On Computing, Communication, Control And Automation (ICCUBEA, Pune, India, 2022, pp. 1-4, doi: 10.1109/ICCUBEA54992.2022.10010733.

[8] Kanase, R. R., Kumavat, A. N., Sinalkar, R. D., Somani, S. (2021). Pose Estimation and Correcting Exercise Posture. In ITM Web of Conferences (Vol. 40, p. 03031). EDP Sciences.

[9] Anilkumar, Ardra, K.T., Athulya, Sajan, Sarath, K.A., Sreeja. (2021). Pose Estimated Yoga Monitoring System. SSRN Electronic Journal. 10.2139/ssrn.3882498.

[10] Irfan, I., Muthalib, M. A. (2022). Implementation of Human Pose Estimation Using Angle Calculation Logic on The Elder of The Hands as A Fitness Repetition. International Journal of Engineering, Science and Information Technology, 2(4), 101-110.

[11] Chen KY, Shin J, Hasan MAM, Liaw JJ, Yuichi O, Tomioka Y. Fitness Movement Types and Completeness Detection Using a Transfer-Learning-Based Deep Neural Network. Sensors (Basel). 2022 Jul 29;22(15):5700. doi: 10.3390/s22155700. PMID: 35957257; PMCID: PMC9371130.

[12] Pauzi, A.S.B. et al. (2021). Movement Estimation Using Mediapipe BlazePose. In: Badioze Zaman, H., et al. Advances in Visual Informatics. IVIC 2021. Lecture Notes in Computer Science(), vol 13051. Springer, Cham. https://doi.org/10.1007/978-3-030-90235-3_49

[13] Ashish Ohri, Shashank Agrawal, Garima S. Chaudhary, "On-device Realtime Pose Estimation & Correction", in International Journal of Advances in Engineering and Management (IJAEM),Volume 3, Issue 7 July 2021.

[14] Long, C., Jo, E. \& Nam, Y. Development of a yoga posture coaching system using an interactive display based on transfer learning. J Supercomput 78, 5269–5284 (2022). https://doi.org/10.1007/s11227-021-04076-w

[15] R. Gadhiya and N. Kalani, "Analysis of Deep Learning Based Pose Estimation Techniques for Locating Landmarks on Human Body Parts," 2021 International Conference on Circuits, Controls and Communications (CCUBE), Bangalore, India, 2021, pp. 1-4, doi: 10.1109/CCUBE53681.2021.9702726.

[16] Bazarevsky, V., Grishchenko, I., Raveendran, K., Zhu, T., Zhang, F., Grundmann, M. (2006). Blazepose: On-device real-time body pose tracking. arXiv 2020. arXiv preprint arXiv:2006.10204.

[17] S. Zhang, W. Chen, C. Chen and Y. Liu, "Human deep squat detection method based on MediaPipe combined with Yolov5 network," 2022 41st Chinese Control Conference (CCC), 2022, pp. 6404-6409, doi: 10.23919/CCC55666.2022.9902631

[18] G. Samhitha, D. S. Rao, C. Rupa, Y. Ekshitha and R. Jaswanthi, "Vyayam: Artificial Intelligence based Bicep Curl Workout Tracking System," 2021 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES), Chennai, India, 2021, pp. 1-5, doi: 10.1109/ICSES52305.2021.9633841.

[19] Jiunn. (2020, April 23). Workout exercise. Kaggle. Retrieved February 25, 2023, from https://www.kaggle.com/datasets/jiunn1998/workout-exercise?select=Original\%2BDatasets

[20] Abdillah, H. (2023, February 18). Workout/exercise images. Kaggle. Retrieved February 25, 2023,from https://www.kaggle.com/datasets/hasyimabdillah/workoutexercises-images?select=biceps\%2Bcurl


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